

## Signals and Systems — Tutorial 6

### System response and stability

### Tutorial information

- It is expected that you attempt the prework questions before coming to the tutorial each week and submit your *handwritten* solutions to your tutor within the first 15 minutes of your tutorial.
- Not all questions are equally difficult. Difficult questions are marked with a ★.
- Questions marked **Extension** go beyond the assessable content for this course and are provided to give a deeper theoretical background and understanding.
- You may not be able to complete all questions within the tutorial time. It is *strongly* recommended that you attempt all questions in your own time and ask your tutor if you are stuck.

### Pework questions

#### A: Impulse Response

Suppose the LTI systems  $y_r(t)$ ,  $y_b(t)$ ,  $y_p(t)$  and  $y_g(t)$  have impulse responses  $h_r(t)$ ,  $h_b(t)$ ,  $h_p(t)$  and  $h_g(t)$  shown in Figure 1. Identify which of the systems are BIBO stable.

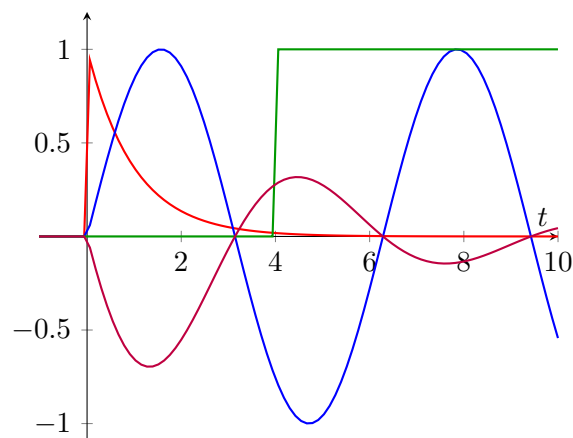


Figure 1: Impulse responses for the LTI systems  $y_r$ ,  $y_b$ ,  $y_p$  and  $y_g$ .

## B: Stability

Let  $\dot{y} = f(u(t), y(t))$  be a LTI system. Recall from Tutorial 1 the system has solutions of the form  $y(t) = Ae^{zt} + B$  which we can plot in the complex plane.

Determine if the systems in the following plots are stable or unstable and if they are stable if they are critically/over-damped or under-damped.

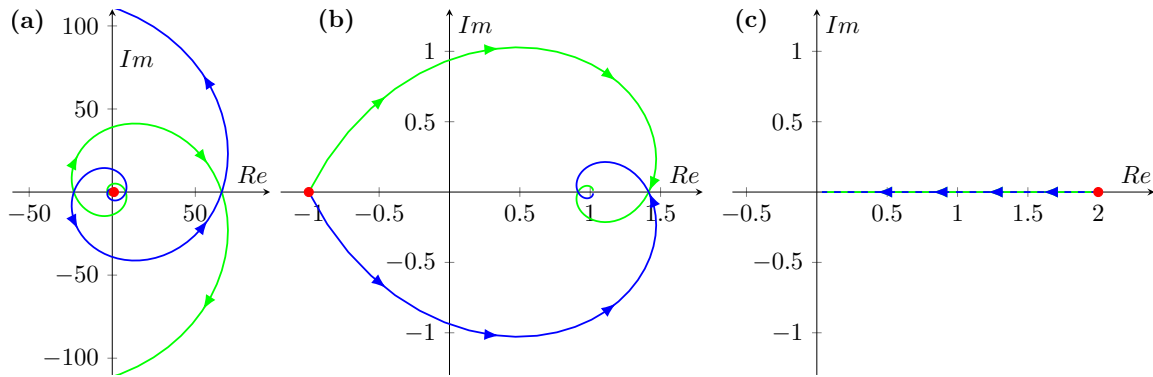


Figure 2: Plots of solutions  $y(t) = Ae^{zt} + B$  where  $A, B \in \mathbb{R}$ ,  $z \in \mathbb{C}$  and  $t \geq 0$ . The blue line is when  $z = x + jy$  and the green is when  $z = x - jy$ . The initial condition is shown by the red point.

## Tutorial questions

### 1: Stability

- Determine if the following systems are stable or unstable. If stable, find the damping coefficient and natural frequency.

(a)  $\dot{x} + 2x = 4u(t)$

(b)  $4\ddot{x} - 2\dot{x} + (5/2)x = u(t)$

(c)  $\ddot{x} + 2\dot{x} + 5x = 5u(t)$

### 2: System Identification and Bode Plots

Three systems  $y_1(t)$ ,  $y_2(t)$  and  $y_3(t)$  have Bode magnitude plots in Figure 3. Identify which of the impulse responses  $h_r(t)$ ,  $h_b(t)$ ,  $h_p(t)$  and  $h_g(t)$  correspond with each of the Bode plots.

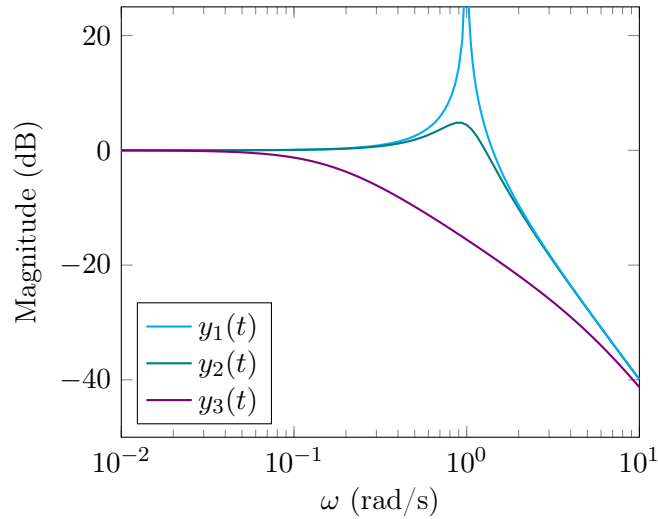


Figure 3: Bode magnitude plots for the systems  $y_1$ ,  $y_2$  and  $y_3$

### 3: Resonance and Type 1 Diabetes

Type one diabetics no longer produce endogenous insulin and rely on the exogenous delivery of insulin to manage and lower blood glucose levels. Typically insulin is delivered via an insulin pump in discrete boluses of size  $A_k$  at some frequency  $\omega_0$  i.e.

$$u(t) = \sum_k A_k \delta(t - kT)$$

where  $T$  is the period. A group of type one diabetics are administered insulin at different frequencies over a twenty-four hour period and the relative increase or decrease of their blood glucose level is recorded. These data are plotted in Figure 4. The total amount of insulin delivered is constant.

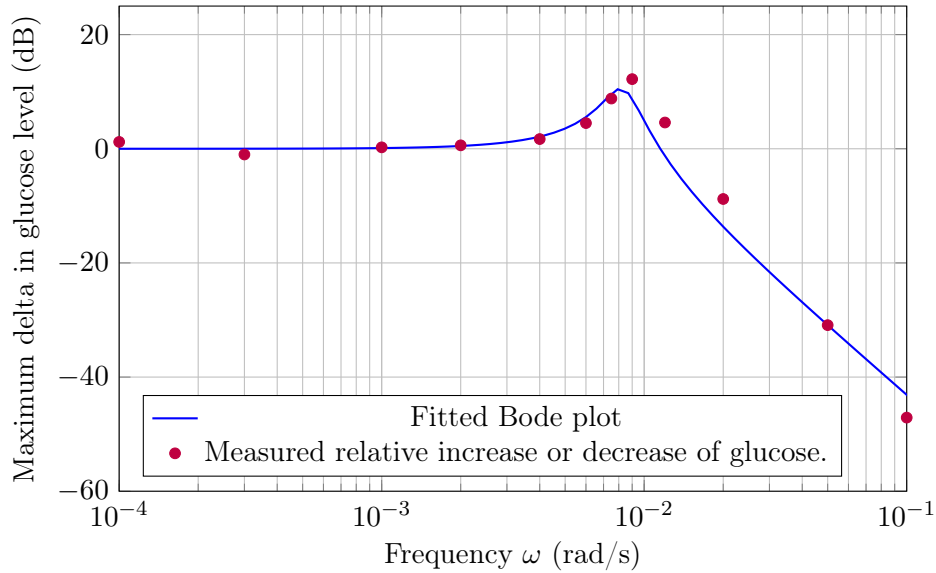


Figure 4: Data and fitted frequency response.

A transfer function

$$H(j\omega) = \frac{G(j\omega)}{U(j\omega)} = \frac{14.4 \times 10^3}{(j\omega)^2 + 36j\omega + 14.4 \times 10^3} \quad (1)$$

is fitted to the data.

1. Is the system BIBO stable? Justify your answer by finding the impulse response and poles of the system.
2. Find the natural frequency  $\omega_n$  and damping ratio  $\zeta$  for the system.
3. Find the  $\omega_0$  which maximises the effectiveness of insulin. What does this  $\omega_0$  represent?
4. Use the inverse Fourier transform to find a model for insulin-glucose dynamics in type one diabetics. How would we identify the optimal frequency to bolus insulin from this time domain model?
5. If this were a model for an RLC series circuit where the input is the voltage of the supply and the output is the voltage across the capacitor give values for the resistor, capacitor and inductor such that transfer function of the circuit matches (1).

#### 4: Chernobyl RBMK Reactor: Thermal Feedback and Stability

The stability of a nuclear reactor is affected by *thermal feedback*. As the reactor temperature changes, the reactivity of the reactor can change, which in turn affects the reactor power. A simplified non-linear model for the thermal power  $P(t)$  and core temperature  $T(t)$  of a nuclear reactor is:

$$\frac{dP}{dt} = (\rho_0 - \beta(T(t) - T_r))P(t) \text{ and } C\frac{dT}{dt} = P - h(T(t) - T_C) \quad (2)$$

which we linearise about the equilibrium, when  $\dot{T} = \dot{P} = 0$ , to give the linear model:

$$\frac{dP}{dt} = -\beta P_0(T - T_0) \text{ and } C\frac{dT}{dt} = P - hT + hT_0 - P_0 = P - h(T - T_C) \quad (3)$$

where

$$T_0 = \rho_0/\beta + T_r \text{ and } P_0 = h(T_0 - T_C)$$

$T_C$  is the coolant temperature,  $T_r = 300^\circ\text{C}$  is a reference temperature,  $\rho_0 = 0.01\text{ s}^{-1}$  is the neutron production at the reference temperature,  $\beta = 1 \times 10^4 (\text{C}^\circ\text{s})^{-1}$  is the thermal feedback rate,  $C = 10\text{ GJ/C}^\circ$  is the thermal capacitance and  $h = 1\text{ MW/C}^\circ$  is the heat feedback.

In the non-linear and linearised models, (2) and (3), the neutron flux  $N \propto P$  and the reactivity of and flux from the control rods is assumed to be constant. Typically the temperature of the coolant is fixed by the ambient temperature of the reservoir used to provide the coolant. We take  $T_C = 290^\circ\text{C}$ .

1. Define  $p(t) = P(t) - P_0$  and  $s(t) = T(t) - T_0$ . Show that (3) is equivalent to

$$\frac{dp}{dt} = -\beta P_0 s(t) \text{ and } C\frac{ds}{dt} = p(t) - hs(t) \quad (4)$$

By eliminating  $p(t)$  show that the transfer function of the system is given by

$$G(s) = \frac{\beta P_0/C}{s^2 + (h/C)s + \beta P_0/C}$$

2. Find the poles for the system (4) and give conditions under which the system is stable or unstable.
3. Assuming the system is stable show (4) has natural frequency  $\omega_n$  and damping ratio  $\zeta$

$$\omega_n = \sqrt{\frac{\beta P_0}{C}} \text{ and } \zeta = \left(\frac{h}{2}\right) \sqrt{\frac{C}{\beta P_0}}$$

4. Find the temperature of the coolant such that the system is critically damped. What happens to the damping ratio  $\zeta$  as  $T_C$  increases if the system is stable?
5. Under certain conditions the RBMK reactor at Chernobyl had a negative thermal feedback rate i.e.  $\beta < 0$ . Under these conditions what happens to the poles of (4)? What would happen if there was a small positive deviation from the equilibrium temperature  $T_0$ ?
6. **[Extension]** Plot  $T(t)$  for  $\beta = -1 \times 10^4 (\text{C}^\circ\text{s})^{-1}$ ,  $0 (\text{C}^\circ\text{s})^{-1}$  and  $1 \times 10^4 (\text{C}^\circ\text{s})^{-1}$ .
7. **[Extension]** If the nuclear reactor were a 1 kg mass on a spring what would be the spring constant ( $k$ ) and coefficient of friction ( $b$ )?
8. **[Extension]** Show that (3) is linearisation of (2).